68.22 TOPS/W Full Boolean Logic Supportable eDRAM-Based Digital CIM with Energy-Efficient Input-Aware Toggle-Rate-Reduced Adder

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***Abstract*—We present an eDRAM-based digital compute-in-memory architecture achieving 68.22 TOPS/W that supports full Boolean logic while enabling energy-efficient computations. This design utilizes a row-sharing full Boolean logic input driver to minimize the area, allowing for Boolean functions and multiply-and-accumulate operations, thereby yielding a multi-computation figure-of-merit (FoM) of 27.1. For energy efficiency in high toggle-rate input activation environments, we implement an input-aware toggle-rate-reduced adder, attaining an efficiency FoM of 4366.08.**

***Keywords*—Artificial intelligence (AI), Boolean logic, compute-in-memory (CIM), embedded DRAM (eDRAM), energy-efficient, toggle-rate**

I. Introduction

Power-constrained edge devices struggle with high energy consumption during large-scale data processing due to excessive data movement [1]. To combat this issue, compute-in-memory (CIM) architectures have gained significant attention, with recent CIM designs demonstrating excellent energy and area efficiencies [2]-[10]. However, as illustrated in Fig. 1 (top), prior CIM macros encounter two primary challenges: 1) CIM architectures specialized for specific operations [2], [3], such as multiply-and-accumulate (MAC), achieve high cell density but lack sufficient flexibility to support various application models. Incorporating additional computational functions to enhance versatility reduces cell density, leading to a trade-off (Fig. 1, top-left) [4], [5]; and 2) In digital CIM (DCIM), most prior works have demonstrated high energy efficiency when the toggle rate of the input activation is below 12.5%. However, in practical convolutional neural networks (CNNs), the toggle rate of the input activation typically exceeds 20% [8], significantly degrading energy efficiency [6] (Fig. 1, top-right). A look-up table (LUT)-based structures [7], [8] can relieve these issues. However, additional circuits for the LUT implementation, such as the 4:1 MUX [7], and the cell area to store the pre-computed value [8] degrade the overall compute density. Furthermore, since the first-stage adder in the LUT-based adder tree [7] is always operated, regardless of the toggle rate of the input activation, consuming a significant amount of power in the process.

To address these challenges, this work proposes a full Boolean logic-supporting embedded DRAM (eDRAM)-DCIM macro (FBL-DCIM) with the following key features: 1) A 4T2C eDRAM macro supports Boolean and MAC operations using a row-sharing full Boolean logic input driver (FBL-ID). This expands the range of supportable computations and applications (e.g., advanced encryption standard (AES), Hamming distance, physical unclonable function (PUF) performance test, Jaccard similarity, and graph data optimization) (Fig. 2, bottom-left) [4], [11]-[14] while minimizing area overhead; and 2)

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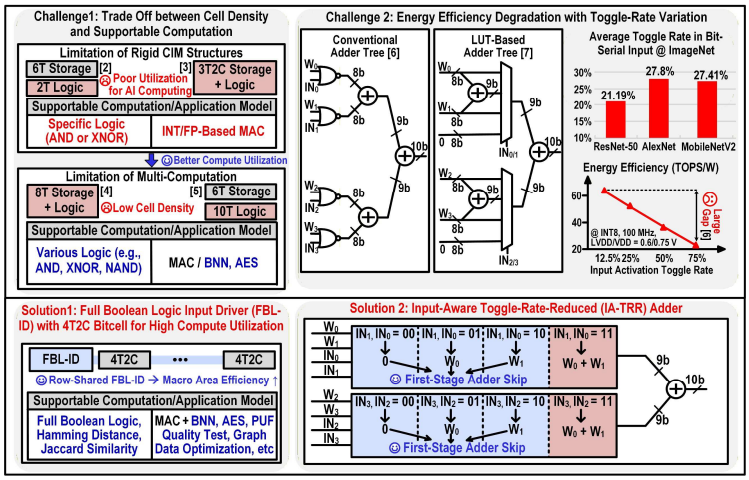


Fig. 1. Two challenges faced by previous compute-in-memory (CIM) macros.

An input-aware toggle-rate-reduced (IA-TRR) adder conditionally bypasses the first-stage adder based on input activation combinations, reducing the average toggle rate of the first-stage adder, improving energy efficiency (Fig. 1, bottom-right).

II. Top Architecture

Fig. 2 shows the overall architecture of the 24-Kb FBL-DCIM macro, divided into 16-Kb MAC- and 8-Kb Boolean logic-specialized macros. Each macro can also operate in memory mode for read/write operations. To minimize power consumption during MAC or Boolean operations, power domains for peripheral and adder circuits are set to VDD and low-VDD (LVDD), respectively.

The MAC-specialized macro executes high-energy-efficiency INT1–8 MAC computations using the IA-TRR MAC compute unit (IA-TRR-MCU). The INT8 MAC operation follows these three stages: 1) Multiplications are performed via a bit-wise AND operation between the 8-bit weight stored in each eDRAM cell and input activation driven by the input driver; 2) Multiplication results are fed into an adder skip circuit (ASC), which selectively operates based on input activation combinations to enhance energy efficiency. The outputs from the ASC are then aggregated into a partial sum (PSUM) using a ripple carry adder (RCA)-based adder tree; and 3) Input activation is supplied bit-serially over eight cycles. Since the accumulator introduces a one-clock delay for accumulating PSUM, the entire operation requires nine cycles to yield the final output.

The Boolean logic-specialized macro supports six operations via the

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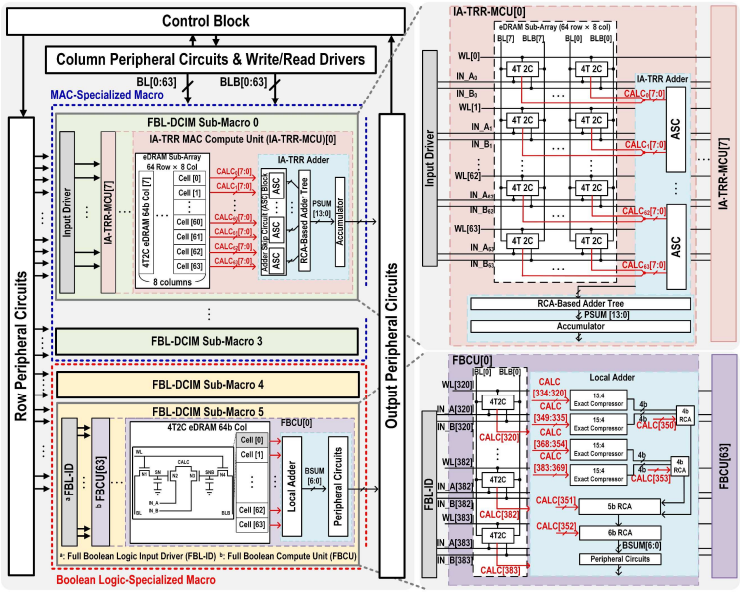


Fig. 2. Overall architecture of the proposed 24-Kb full Boolean logic-supporting embedded DRAM-digital CIM (FBL-DCIM) macro.

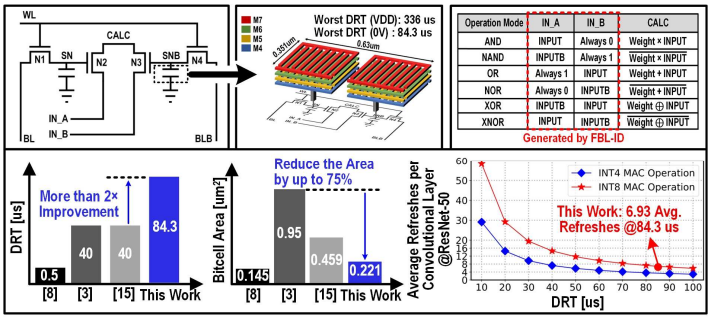
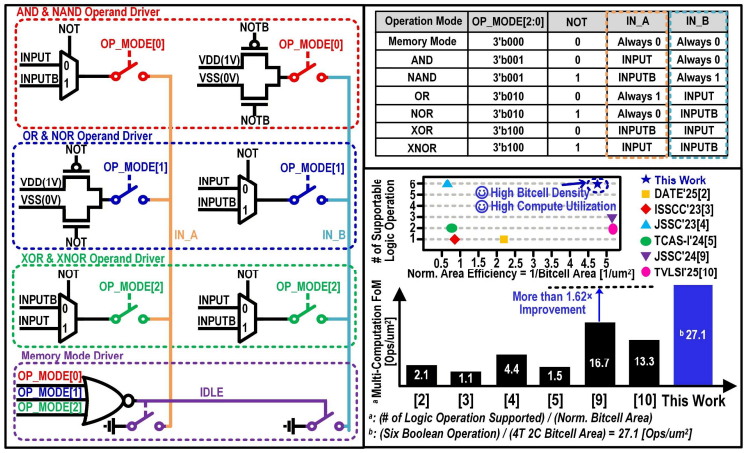


Fig. 3. 4T2C eDRAM cell implementation (top left), operand table (top right), and performance improvements (bottom).

FBL-ID. The Boolean operation outputs (CALC) from each cell in each column are summed into a 7-bit output (BSUM [6:0]) using local adders and sent to external logic via peripheral circuits [10]. Several CIM structures have enhanced Hamming distance stability at low voltages [11], integrated PUF-based security [12], accelerated computationally challenging Jaccard similarity tasks [13], and optimized graph convolution network sparse-access efficiency [14]. Therefore, the versatile FBL-DCIM macro can unify these varied computational capabilities within a single, efficient memory architecture.

III. 4T2C eDRAM Cell with Full Boolean Logic Input Driver

As shown in Fig. 3 (top left), the 4T2C eDRAM cell receives two input signals (IN_A and IN_B) generated by the FBL-ID based on the selected operation mode. The limited data retention time (DRT) of the eDRAM cell poses challenges for handling multiple operations within a single macro, requiring either large cell capacitors or frequent refresh operations. To address this, we stack a customized metal–oxide–metal capacitor (MOMCAP) onto the bitcell (Fig. 3, top center). Since the MOMCAP is designed to fit within the cell layout, we can achieve up to a 75% reduction in bitcell area compared to prior eDRAM CIM cells [3], [8], [15] (Fig. 3, bottom center). The worst-case DRT of the 4T2C cell is 84.3 us, representing at least a 2× improvement over previous studies [3], [8], [15] (Fig. 3, bottom left). This DRT allows an average



Operation Mode	OP_MODE[2:0]	NOT	IN_A	IN_B
Memory Mode	3'b000	0	Always 0	Always 0
AND	3'b001	0	INPUT	Always 0
NAND	3'b001	1	INPUTB	Always 1
OR	3'b010	0	Always 1	INPUT
NOR	3'b010	1	Always 0	INPUTB
XOR	3'b100	0	INPUTB	INPUT
XNOR	3'b100	1	INPUT	INPUTB

Fig. 4. Full Boolean logic input driver (FBL-ID) implementation (left), operand table (top right), compute utilization gap (right), and performance improvement (bottom right).

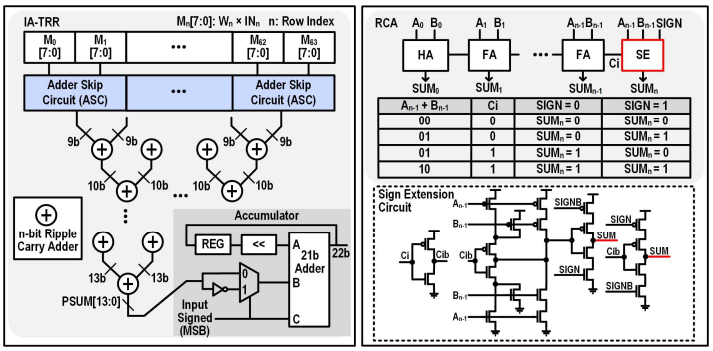


Fig. 5. Structures of the input-aware toggle-rate-reduced (IA-TRR) adder (left) and ripple carry adder (RCA) (right).

of only 6.93 refreshes per convolutional layer for ResNet-50 using the ImageNet data set, assuming that the single eDRAM CIM macro is used in INT8 MAC computation mode (Fig. 3, bottom right). Therefore, macro stacking the MOMCAP is well-suited for system-level implementations.

To support full Boolean computation while minimizing area and power at the system level, we propose a row-sharing FBL-ID architecture controlled by the operation mode signal (OP_MODE[2:0]). Fig. 4 shows the FBL-ID implementation, operand table, and computation performance improvements. A NOT signal is utilized to execute NAND, NOR, and XNOR operations without increasing the OP_MODE's bit-width. The FBL-ID consists of four 2:1 multiplexers, eighteen switch transistors, and a single 3-input NOR gate. As illustrated in Fig. 4 (right), proposed DCIM macro achieves high compute utilization while maintaining high bitcell density, even while supporting a large number of logic operations. The row-sharing FBL-ID allows the proposed macro to perform full Boolean operations while achieving the best multi-computation figure of merit (FoM) with minimal overhead in the bitcell area. This FoM, representing area efficiency across a wide range of operations, is more than 1.62× higher than that of prior works [2]-[5], [9], [10].

IV. Input-Aware Toggle-Rate-Reduced Adder

Fig. 5 illustrates the proposed IA-TRR adder and RCA. The adder supports reconfigurable operations across different bit widths (1-8 bits)

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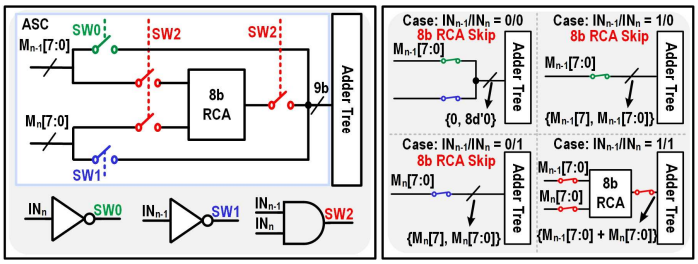


Fig. 6. Structure (left) and operation (right) of the adder skip circuit (ASC).

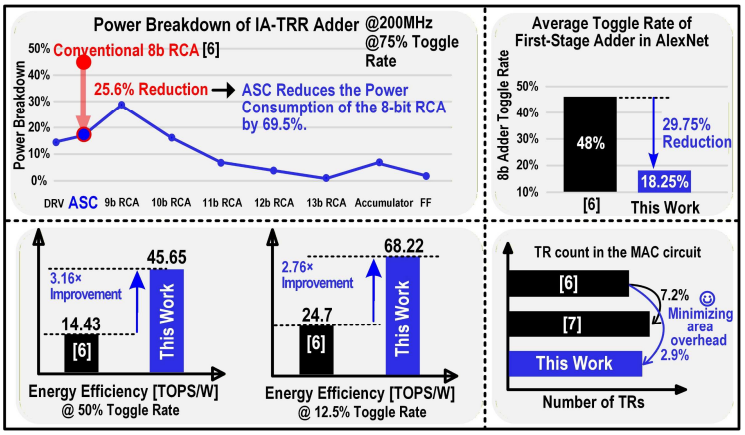


Fig. 7. Performance improvements owing to the proposed IA-TRR adder structure.

and signed or unsigned weights and input activations. The signed extension is activated in the RCA when signed weights are used.

Fig. 6 shows the block diagram and operation of the ASC. The ASC core consists solely of an 8-bit RCA and control switches, minimizing area overhead. Its operation is conditionally controlled by the adjacent input activations, IN_{n-1} and IN_n. If either IN_{n-1} or IN_n is 0, the control switches prevent data from entering the 8-bit RCA. Both the proposed ASC and the LUT-based adder [7] have the common goal of reducing unnecessary computations. However, the ASC architecture adds switches between the bitcell and the 8-bit RCA, restricting unnecessary 8-bit RCA operations based on input activation patterns. In contrast, the previous LUT-based adder [7] must precompute all possible addition results in the 8-bit RCA whenever the weight changes, degrading energy efficiency. Furthermore, when both IN_{n-1} and IN_n are "0," the LUT-based adder [7] connects VSS to the MUX input and outputs that signal. In contrast, the proposed ASC architecture eliminates the need for an explicit VSS connection by directly passing M_{n-1} or M_n to the adder tree, thereby reducing unnecessary circuit overhead and simplifying the overall structure. An additional advantage of this architecture is its ability to retain the previous 8-bit RCA output. Input activations are provided in a bit-serial format over eight cycles. When both IN_n and IN_{n-1} become 1 for the first time, the 8-bit RCA computes the result ($M_{n-1} + M_n$) and holds it until the MAC operation is completed. For other input activations, the result is maintained in a floating state at the 8-bit RCA output node, with the charge retained by its parasitic capacitance of approximately 4 fF. Since retention is needed for only 8 cycles in the worst case, the voltage drop from leakage is negligible, ensuring reliable data retention. Therefore, if IN_{n-1} and IN_n are 1 in a later cycle, the 8-bit RCA avoids unnecessary recalculations since it remembers the previous result, significantly reducing transitions and power consumption.

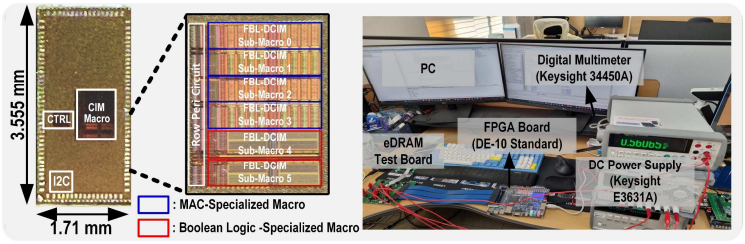


Fig. 8. Die photograph of the prototype chip (left) and measurement setup (right).

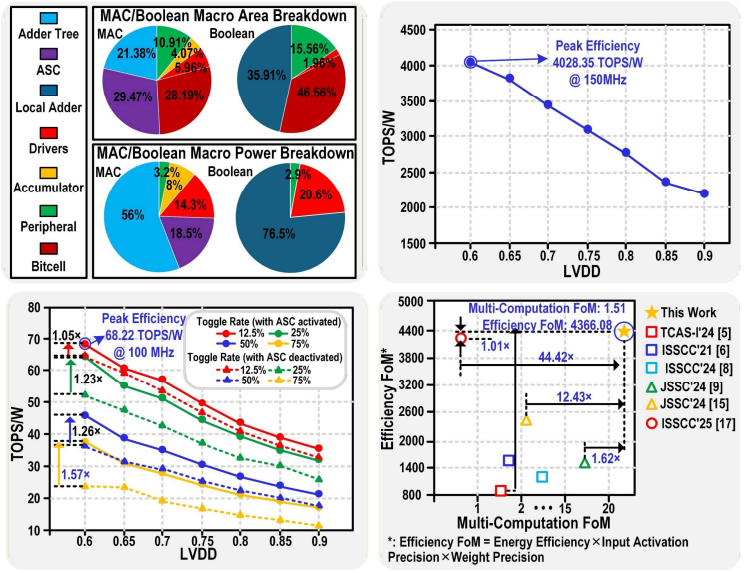


Fig. 9. Macro power and area breakdown (top left), energy efficiency of INT1 BOOL (top right) and INT8 MAC (bottom left) operations, and position chart to compare the efficiency and multi-computation FoMs with prior designs (bottom right).

Fig. 7 illustrates performance enhancements with the proposed IA-TRR adder structure. In the INT8 MAC operation, the ASC reduces the power consumption of the 8-bit RCA by 69.5% at 200 MHz with a 75% input activation toggle rate, decreasing the power contribution of the 8-bit adder from 44.4% to 18.8%. This is because the ASC reduces the average toggle rate of the first-stage adder from 48% to 18.25% in the AlexNet. At a 12.5% toggle rate, the proposed circuit achieves 2.76× higher energy efficiency than the conventional structure. Furthermore, at a 50% toggle rate, the proposed circuit achieves up to 3.16× higher energy efficiency than the previous structure, demonstrating superior performance under high toggle-rate conditions. Furthermore, while achieving high energy efficiency, the number of transistors required for MAC operations increases by only 2.9% compared to the conventional structure [6], minimizing the area overhead.

V. Measurement Results

Fig. 8 shows the die micrograph of the proposed FBL-DCIM macro implemented in a 28-nm CMOS process, along with the measurement setup and performance summary.

Fig. 9 presents the performances of the proposed FBL-DCIM macro. Power and area breakdown results show that the first-stage adder consumes over 18.5% of the total power using the IA-TRR structure (Fig. 9, top left). In the INT1 BOOL operation, peak energy efficiency reaches 4028.35 TOPS/W at LVDD = 0.6 V and 150 MHz (Fig. 9, top right). In the INT8 MAC operation, the measured energy efficiency

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SRAM-Based CIMs → Digital CIMs; eDRAM-Based CIMs → Analog CIMs

	TCAS-I'24 [5]	S.VLSI'25 [16]	S.VLSI'22 [7]	This Work	ISSCC'24 [8]	JSSC'24 [9]	JSSC'24 [15]
Technology	55nm CMOS	28nm CMOS	12nm CMOS	28nm CMOS	28nm HKMG	40nm CMOS	28nm CMOS
Multi Boolean Operation	No (XOR/AND)	No	No	Yes (AND/NAND/OR/NOR/XOR/XNOR)	No	Yes (NAND/NOR/XOR)	No
Bit Cell Structure	6T + 10T	6T	6T	4T2C	3T	3T	3T2C
Array Size	4 Kb	12 Kb	8Kb	16 Kb ([d] MSM) / 8Kb ([e] BLSM)	40 Kb	144 Kb	800 Kb
Macro Size [mm²]	0.062	1.56	0.032	0.05 (MSM) / 0.02 (BLSM)	0.017	0.088	2.03
Bit Cell Area [um²]	[a] 1.09	-	-	0.221	0.145	[a] 0.181	0.459
DRT [us]	N/A	N/A	N/A	84.3-336	0.5-1.3	3.7 (@ 300 K)	40
Supply Voltage [V]	0.7-1.2	0.6-1	0.72	0.75-1 (VDD) / 0.6-0.9 (LVDD)	0.9-1.2	1.1	1
Frequency [MHz]	100	85-410	-	50-250 (MSM, BLSM)	400-1100	180-340	250
Input/Weight Precision [b]	1-8 / 1-8	4, 8 / 4, 8	4-8 / 4, 8	1-8 / 1-8	1-8 / 8	4 / 4	8 / 9
Output Precision [b]	7-22	-	18	7 (INT1), 22 (INT8)	22	-	-
Full Precision	Yes	Yes	Yes	Yes	Yes	No	No
Energy Efficiency [TOPS/W] (Weight 50% Sparsity)	2238 (INT1), 8 (50% [b] ITR, INT8) 14.2 (10% ITR, INT8)	36.3 (INT8)	30.3 (10% [f] ISP , INT8)	4028.35 (INT1, BLSM), 37.5 (75% ITR, INT8, MSM) 68.22 (12.5% ITR, INT8, MSM)	13 (50% [f] ISP, INT8) 18.1 (10% ISP, INT8)	92.5 (INT4, @ 300 K)	33.9-40.6 (8b-9b)
Compute Density [TOPS/mm²]	[a] 50.75 (INT1) [a] 0.81 (INT8)	0.41 (INT8)	[a] 1.28 (INT8)	204.8 (INT1, BLSM) 2.28 (INT8, MSM)	16.2 (INT8)	[a] 580 (INT4, @ 4.7 K)	0.72 (8b-9b)
Throughput [GOPS]	[c] 1638.4 (INT1) [c] 51.2 (INT8)	[c] 837.9 (INT8)	[c] 672 (INT8)	[c] 4096 (INT1, BLSM) [c] 113.8 (INT8, MSM)	[c] 80 (INT8)	-	[c] 11.2 (8b-9b)
Efficiency FoM	908.8	2323.2	1939.2	4366.08	1158.4	1480	2440.8
Multi-Computation FoM	1.5	-	-	27.1	6.89	16.7	2.18

[a]: Normalized to 28nm, [b]: Input Toggle Rate, [c]: Normalized to 16Kb (INT8) and 8Kb (INT1),
[d]: MAC-Specialized Macro, [e]: Boolean Logic-Specialized Macro [f]: ISP = Input Sparsity

Fig. 10. Performance comparison of state-of-the-art SRAM- and eDRAM architectures for ACIM and DCIM.

increases as LVDD decreases (Fig. 9, bottom left). When the ASC circuit is activated, with 50% weight sparsity and an input activation toggle rate of 12.5–75%, the energy efficiency ranges from 37.49 to 68.22 TOPS/W at LVDD = 0.6 V and 100 MHz, showing superior performance compared to the case when the ASC circuit is deactivated across all toggle rates. Specifically, at a 75% toggle rate, the energy efficiency improves by up to 1.57×. This demonstrates the superiority of the ASC circuit. Furthermore, the efficiency FoM, defined as the product of energy efficiency, input activation precision, and weight precision, improves by 1.01–5.34× compared to prior works, while the multi-computation FoM, defined as the ratio of the number of supportable computations to the bitcell area for multi-computation, improves by 1.62–12.43× over prior eDRAM CIM designs.

Fig. 10 compares the performance of the proposed FBL-DCIM macro against existing state-of-the-art CIMs. Unlike prior DCIM and ACIM designs, the proposed macro enhances computational versatility by supporting six Boolean operations alongside MAC and memory operations. Utilizing a novel row-sharing FBL-ID architecture, it achieves the best multi-computation FoM of 27.1, demonstrating remarkable area efficiency for applications demanding operational flexibility. At VDD = 0.75 V, LVDD = 0.6 V, and 100 MHz, this architecture reaches a peak energy efficiency of 68.22 TOPS/W, marking an improvement of up to 4.81× over prior DCIMs. With an average toggle rate of 25% in practical models, our DCIM achieves a performance level of 64.24 TOPS/W, translating to a 5.35× enhancement in energy efficiency over prior designs. Moreover, this work is more energy-efficient than prior arts [7], [8], even though they report under more favorable high-sparsity conditions, which are less demanding than our 75% toggle rate. Unlike eDRAM-based ACIMs that sacrifice accuracy to improve energy efficiency, the proposed architecture maintains full precision while offering higher energy efficiency.

VI. Conclusion

This paper presents a 68.22 TOPS/W eDRAM-DCIM in a 28 nm process. A row-shared FBL-ID is applied to the 4T2C eDRAM cell, supporting both MAC and full Boolean logic operations with high area efficiency, achieving a 27.1 multi-computation FoM. Futhermore, an IA-TRR adder conditionally bypasses its first-stage adder computation during INT8 operation, improving energy efficiency by up to 1.57× at high toggle rates.

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